



Vortex-excited force evolutionary characteristics of split three-box girder bridges during vortex-induced vibration

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ABSTRACT

As novel cross sections, aerodynamic behaviors and vortex-induced vibration (VIV) inducement of split box girders, especially split three-box girders, are not fully understood. Analysis of vortex-excited force (VEF) evolutionary characteristics in the VIV process may reveal valuable information. Based on displacement and pressure measurement wind tunnel tests on a large-scale section model, the evolutionary characteristics of typical VEF indexes were expounded. VIV was divided into four stages: pre-VIV, ascent, extreme point, and descent. At the extreme point, high-order harmonics were found, which indicates that VEF has obvious non-linearities. Mean and RMS values of global VEF appear to be controlled by upstream and downstream boxes. An increase in RMS values is the main cause of VIV. VEF spanwise correlation is mainly dominated by structural vibration at the extreme point, while spanwise distance is dominant in other stages. Local aerodynamic forces on the surfaces include upstream box surface (excluding partial lower surfaces of the wind fairing), upstream and downstream curved surfaces in upstream gap, upper surface of downstream box, and the wake are the main contributors to global VEF and the main cause of VIV. This work provides a comprehensive understanding of aerodynamic behaviors and VIV inducement of split three-box girder bridges.

1. Introduction

Advancements in design theory and construction technology have made modern bridges longer in span and more flexible than ever before. The structural stiffness, natural frequency, and torsion-bending ratio become smaller as the bridge span and flexibility increase, which makes the bridge more sensitive to wind loads. Wind resistance is currently the primary concern in the design of long span bridges. Wind-induced vibration forms generally include galloping, flutter, buffeting and VIV. Galloping is characterized by pure bending and large-scale vibration caused by the self-excitation of airflow. Slender structures, such as cables, are most susceptible to galloping. Flutter is a self-excited and divergent vibration. Once occurs, it may cause devastating damage to structures and must be avoided in the wind-resistant design. Buffeting is a random vibration in turbulent flows. VIV is a self-excited limited-amplitude forced vibration that occurs under uniform or small turbulent flows. As air flows through the surface of bluff bodies, it generates periodic vortex shedding around the surface and induces VIV as the shedding frequency approaches the natural frequency of the structures. VIVs occur at relatively low wind speeds where there is a high

probability of occurrence at bridge site, which is likely to cause problems such as driving safety and structural fatigue.

Split box girders are commonly used in long span cable-bearing system bridges to satisfy the demand for traffic volume and flutter stability performance, especially in coastal areas where extreme weather makes the design benchmark wind speed particularly high. However, their aerodynamic behavior is not yet fully understood and VIV control measures are still under investigation. The split box girder has excellent flutter performance, but shows various degrees of VIV compared to the streamlined box girder due to the existence of gaps (Hirano et al., 2001). Vertical VIVs are common in twin box girders (Yang and Ge, 2012; Falbe-Hansen et al., 2004; Li et al., 2018). The gaps have a certain distance from the centroid and the bridge deck is relatively wide. Split three-box girders tend to experience vertical and torsional VIVs (Brancaleoni and Diana, 1993; Diana et al., 2013). However, a wider bridge deck with more boxes can satisfy the traffic needs of different vehicle types, which is undoubtedly an ideal girder form for long-span highway and railway bridges.

VIV research methods mainly include field measurement, wind tunnel test, numerical simulation, and mathematically analytical

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modeling. Despite the high cost and poor controllability, some scholars still use field measurements to research VIV (Li et al., 2011; Ma et al., 2020; Zhu et al., 2018; Cao et al., 2020). Numerical simulation and mathematically analytical modeling generally require additional verifications due to inherent limitations and a relatively small scope of application (Larsen and Walther, 1997; Xu et al., 2017; Li and Laima, 2015). As for now, wind tunnel tests are the most commonly used and reliable approach (Yang and Zhou, 2014; Ma et al., 2018). Thanks to flow visualization technology, wind tunnel tests are more superior in revealing VIV mechanism (Flemming and Williamson, 2005; Yuan et al., 2017).

VIV control methods generally include structural measures, mechanical measures, and aerodynamic measures. Aerodynamic measures, the most widely used among them, are economical, convenient, and reflect the significance of aerodynamic optimization. The mechanism must first be clarified in seeking any appropriate countermeasures to suppress aerodynamic responses; VEF evolutionary characteristics in the VIV process must be understood before exploring the VIV inducement mechanism.

Many scholars have previously explored VEF. Diana et al. (2006), for example, investigated the vortex shedding phenomenon during VIV in the Messina Strait bridge to find that vortex shedding or VEF is closely related to VIV amplitude. A single degree of freedom nonlinear time-domain mathematical model that can reproduce VIV characteristics with only one or more modal shapes was proposed as a tool for predicting whether a structure is affected by VIV. Based on numerical simulation and wind tunnel tests, Chen et al. (2020) studied the vertical VIV of a flat-closed-box girder. The distribution of vortex-induced pressures and energy, and their evolution law was discussed. They pointed out that the linear aerodynamic negative damping and nonlinear aerodynamic positive damping are the reason for the rapid development of VIV. The main contributors to positive and negative aerodynamic damping were clarified. Wind tunnel tests of a cylinder with low damping, large mass ratio, and high Reynolds number can obtain VIV amplitudes larger than its diameter. The results show that there is a significant correlation between the wake and the aerodynamic forces, which corresponds to the vortex shedding switching from one mode to another. Thus, wake harmonic analysis can reveal the vortex shedding mode when test conditions do not allow pressure measurement or force measurement, highlighting the importance of wake control in VIV control (Belloli et al., 2012). Pressure and displacement measurement technology can be deployed simultaneously to obtain the surface pressure distribution and VIV displacement of a streamlined box, which reveal the evolutionary law of aerodynamic force throughout the VIV process. The downstream upper surface of the box makes a major contribution to VIV. The installation of a 30° inclination spoiler can compress the air flowing through the bridge deck, change the separation and reattachment position of the vortex, and suppress VIV (Hu et al., 2018).

The VEF spanwise correlation also plays an important role in VIV process. Adding a spanwise sinusoidal perturbation module to the front and back ends of the section model can weaken the VEF spanwise correlation to optimize VIV performance. Pressure measurements show that the spanwise sinusoidal perturbation module can stabilize the attached flow, eliminating the shedding peak of the fluctuating pressure component under high wave steepness (El-Gammal et al., 2007). Compared with the uniform flow, turbulence can restrain the formation of vortex and thus restrain VIV. Daniels et al. (2016) used large eddy simulation (LES) to analyze the flow field around an elastically mounted rectangular cylinder with an aspect ratio of 4 in smooth and free turbulent flows. The results show that the amplitude of oscillation decreases obviously with the increase of turbulence intensity. The increase of turbulence integral scale will increase the vibration amplitude because it enhances the spanwise correlation of surface pressures. The flow around the split box girder is more complicated than in other bridge sections due to the existence of the gaps. The flow alternates up and

down within the gaps introducing a periodic lift or moment that cause VIVs. For this reason, VIV control in split box girders is often targeted towards the gaps (Yang et al., 2016).

At present, research on fluid-structure interaction (FSI) in VIV process mainly centers on semi-empirical mathematical models of the relationship between VEF and the VIV amplitude. There have been relatively few studies on the evolutionary characteristics of local aerodynamic forces relative to global VEF in the VIV process, and even fewer in regards to novel split three-box girders. In the present study, wind tunnel tests were conducted to gather large-scale section model displacement and pressure measurement information for a split three-box girder bridge. The displacements and pressure time-histories throughout VIV were obtained accordingly. Synchronous displacement and pressure measurement technology revealed the evolutionary characteristics of typical VEF indexes (including components, spanwise correlations, mean and RMS values of wind pressure coefficients, correlations between local aerodynamic forces and global VEF, contributions of local aerodynamic forces to global VEF, and phase lags between local aerodynamic forces and global VEF at predominant frequency) in the VIV process. VIV was divided into four stages: pre-VIV, ascent, extreme point, and descent. The results of this work may provide in-depth understanding of aerodynamic characteristics, VIV inducement mechanism and purposeful selection of VIV control measures in split three-box girder bridges.

2. Background and experimental setup

A long span highway-railway split three-box girder cable-stayed and suspension cooperative system bridge has a main span of 1488 m, a total length of 2664 m, and a tower height of 294 m. The mid-span main cable has a rise-span ratio of 0.154, suspension to span ratio of 0.45, and nine pairs of cross cables. The pure suspension area at the mid-span is 452 m. The girder consists of three separate streamlined boxes 5 m in height. The middle railway box and highway boxes on both sides are 13 m and 16.7 m in width, respectively. The gap width is 8.05 m and the wind fairing width is 2.75 m. A box cross beam with a width of 2.8 m and a spacing of 14 m connects the three boxes to ensure they have coordinated force. The layout and cross section of this structure is shown in Fig. 1.

The experiment was conducted in XNJD—3 boundary layer wind tunnel with a test section 36 m in length, 22.5 m in width, and 4.5 m in height, where wind speed ranges from 1.0 m/s to 16.5 m/s. The blockage ratio is approximately 4%. The large-scale section model (scale ratio of 1:30) is 3.9 m in length, 2.267 m in width, and 0.167 m in height. The main body and bridge deck of the model are comprised of high-quality wood. The pressure measurement sections are composed of phenol board with smooth surface and good plasticity. The safety barrier and maintenance tracks were fabricated from engineering plastics (Fig. 2). The distances between sections from A to D are 370 mm, 190 mm, and 800 mm, respectively (Fig. 3a). 90 tubes are arranged on each pressure measurement section (Fig. 3b). The minimum and maximum distances between the tubes are 20 mm and 60 mm, respectively. The tubes are encrypted on the transition curved surface of the lower surface.

The test setup was designed specifically for dynamic tests on bridge sections and was mounted on a workable frame. The dynamic model was suspended by four pairs of linear springs and allowed to vibrate vertically and torsionally. To prevent the test support system from impacting the uniform flow, the front and rear ends of the frame were equipped with diversion wind fairings. Free vibration was used to synchronously obtain the displacement and wind pressure time-histories in the test. Two non-contact *Micro-Epsilon* laser displacement sensors were symmetrically installed at the ends of the model end plates to measure the vertical and torsional displacements; a *DSM4000* Scanivalve System was adopted for pressure measurement, as shown in Fig. 4. Scanivalves were installed inside the model to avoid disturbing the aerodynamic shape and flow. The scanivalves have short aerodynamic connections to the

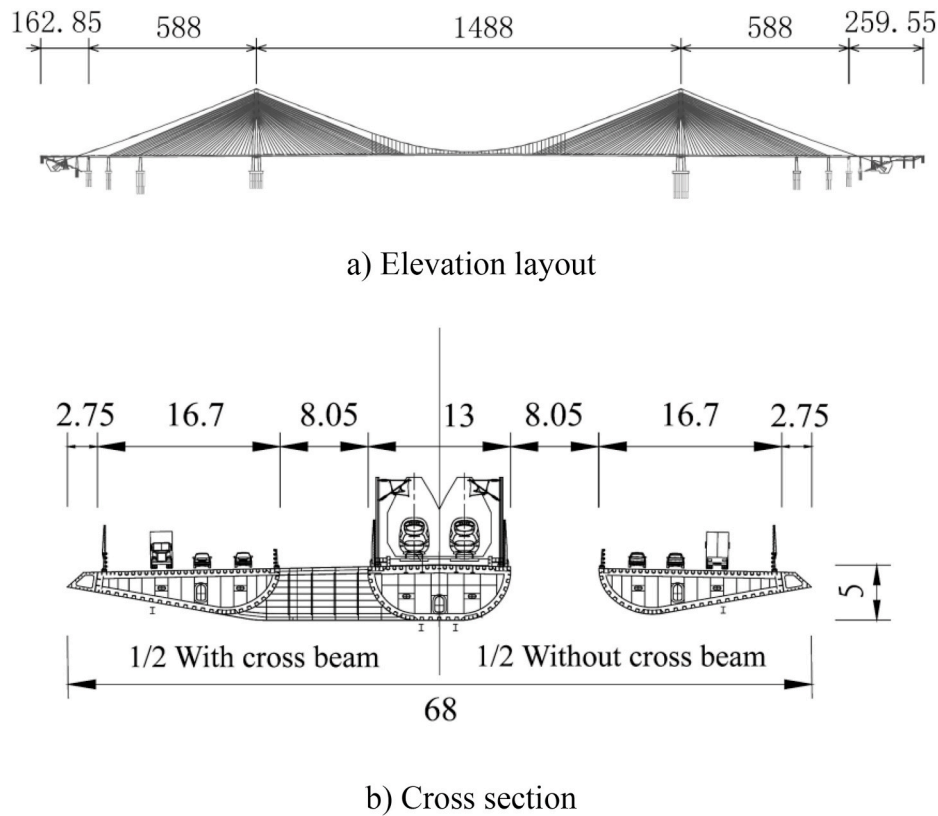


Fig. 1. Elevation layout and cross section diagram of the bridge (unit: m).

tubes that ensure the high-frequency response of the system. The sampling frequency is 256 Hz, which is much higher than the structural natural frequency. The sampling time is 60 s and the number of sampling points is 15 360. The length of the tubes on one section is the same, and the maximum tube length is 700 mm. The pressure signals are corrected by the frequency response function of the tubing system, summarized from Irwin et al. (1979) as shown in Fig. 5. The amplitude ratio represents the ratio of the real amplitude to the measured amplitude. The phase lag represents the time delay between real and measured signals.

The system dynamic parameters listed in Table 1 were identified by free vibration method before the test. The amplitude ratio and phase lag at the first-order torsional frequency of 3.39 Hz are 1.02 and -5.2° , indicating that the difference between real and measured signals is very small. The mass damping parameter (Scruton number) $Sc = 4\pi I \xi / \rho B^2 D^2$ (where I is the mass moment of inertia per unit length, ξ is the damping ratio of the structure, ρ is the air density with a value of 1.225 kg/m^3 , and B and D are the width and depth of the section, respectively) in the torsional mode is 5.75, which is smaller than the target value of 6.79. A smaller Sc makes for larger VIV amplitudes and wider lock-in regions.

3. VIV response and means of analysis

The test wind speed varies from 1 m/s to 9 m/s. The corresponding Re (Reynolds number) range is 1.16×10^4 to 1.05×10^5 . The Reynolds number $Re = UD/\nu$ (where U is the incoming wind speed, D is the depth of the section, and ν is the kinematic viscosity) represents the ratio of inertial force to viscous force, and is a decisive parameter in the transition from laminar to turbulent flow. When Re is large, inertia effect plays a major role; when Re is small, the viscous effect is stronger. Limited by size of the wind tunnel and test equipments, the Re number is generally hundreds or even thousands of times smaller than the real-bridge value. However, the wind resistance design of the bridge under low Re is usually conservative (Schewe, 2001).

Fig. 6 shows the VIV response under uniform flow. There are two

torsional VIV lock-in regions at the 0° attack angle in the completion state. The corresponding Strouhal number $St = fD/U$ (f is the vortex shedding frequency) ranges are 0.067–0.090 and 0.107 to 0.139, respectively. In the optimized case (with 30% ventilation ratio grids in the gaps and a new maintenance tracks scheme), both torsional VIV regions disappear. A similar phenomenon was also found and reported in Diana's (2013) research.

This study centers on the VEF evolutionary characteristics of split three-box girder bridges under a specific VIV pattern rather than distinguishing differences between the first and second VIV lock-in regions or VIV control measures. With the second torsional VIV region as the research object, the VIV process was divided into four stages: pre-VIV, ascent, extreme point, and descent (marked with different shapes in Fig. 6). The evolutionary characteristics of VEF were analyzed accordingly. The reduced wind speeds $U^* (U^* = U/f_t B)$, where f_t is the first-order symmetrical torsional frequency of the model) corresponding to the four stages are 0.71–0.81, 0.85 to 0.95, 0.97, and 0.99 to 1.10, respectively. The torsional angle in the pre-VIV stage is almost zero. In the ascent stage, the wind-induced torsional angle gradually increases to the extreme point (0.446°) as wind speed increases. In the descent stage, the torsional angle decreases sharply to zero as wind speed increases.

Reduced wind speeds of 0.80, 0.90, 0.97, and 1.04 were selected as the representative wind speeds of the four stages with corresponding torsional amplitudes of 0.004° , 0.280° , 0.446° , and 0.109° . Fig. 7a shows the torsional angle time-history of the model at the extreme point; Fig. 7b shows the torsional angle displacement spectra of the four stages normalized by the maximum value among them. The predominant frequency of the spectrum in pre-VIV is 3.41 Hz with a small normalized amplitude, indicating that the model has only slight vibrations. The predominant frequency in ascent stage is identical to the natural frequency of the model; predominant frequencies of the extreme point and descent stage are very close to the torsional natural frequency of the model (3.39 Hz), but slightly larger as the normalized amplitude reaches the maximum at the extreme point. The predominant frequencies of the

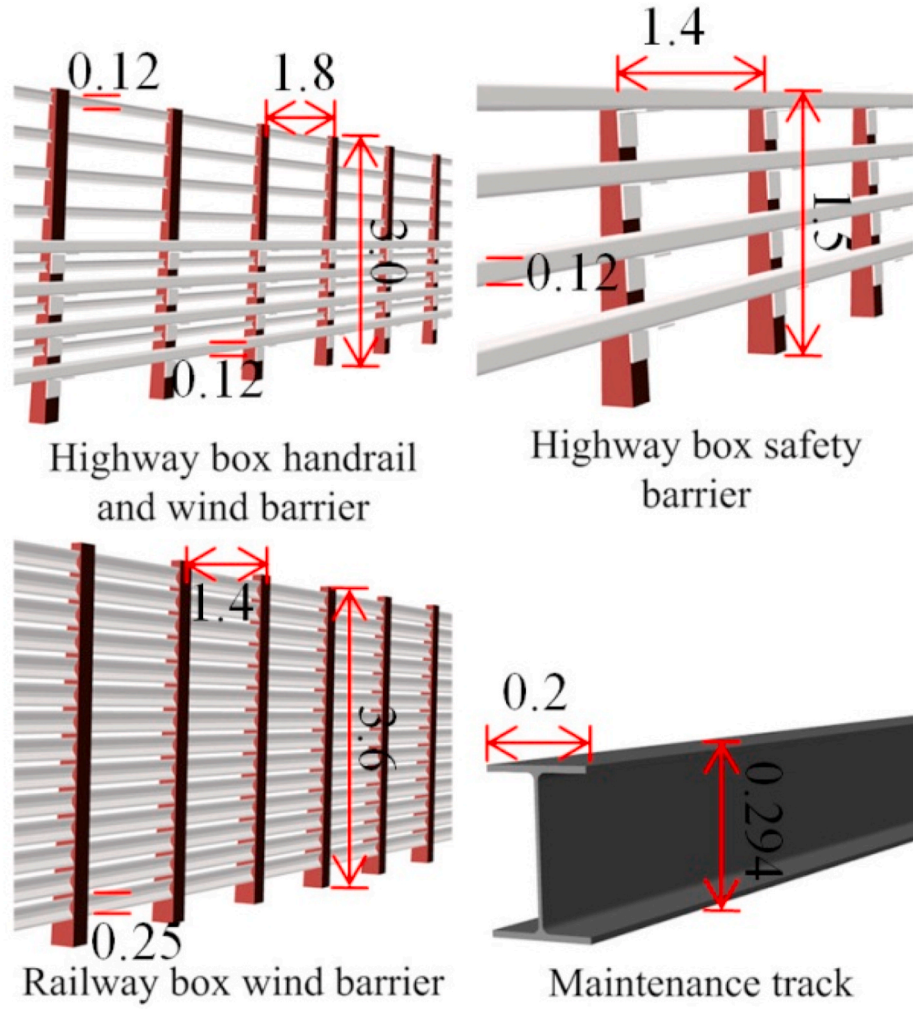


Fig. 2. Ancillary facilities (unit: m).

four stages are not identical. The torsional angle displacements and their predominant frequencies show obvious evolutionary characteristics in the VIV process.

4. Vortex-excited force

Synchronous pressure and displacement measurement technology facilitates analysis of VEF evolutionary characteristics in different VIV stages or under different VIV amplitudes. The VEF can be obtained by integrating the pressure of local tubes along the cross section. For comparison among pressure values with different wind speeds, the VEF coefficient was obtained by dividing VEF by the dynamic pressure $0.5\rho U^2$, as shown in Eqs. (1)–(3). $C_L(t)$, $C_D(t)$ and $C_M(t)$ are the total lift, drag, and moment coefficient time-histories per unit length, respectively. $p_i(t)$ is the fluctuating wind pressure time-histories of the i -th tube. n is the total number of the tubes. ΔL_i is the average coverage area of the i -th tube. α_i is the angle between the normal of the i -th tube and the section surface. D is the depth of the section. x_i and y_i are the arms of forces of the i -th tube relative to the section centroid. This study centers on torsional VIV, so the VEF analysis given in this section focuses on $C_M(t)$.

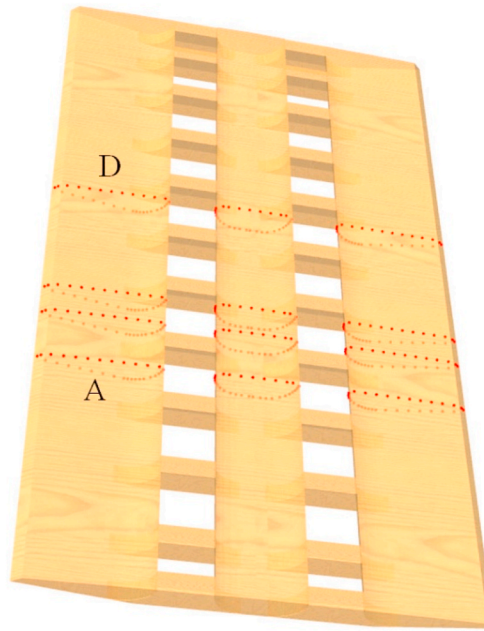
$$C_L(t) = \frac{\sum_{i=1}^n p_i(t) \Delta L_i \cos(\alpha_i)}{0.5\rho U^2 D} \quad (1)$$

$$C_D(t) = \frac{\sum_{i=1}^n p_i(t) \Delta L_i \sin(\alpha_i)}{0.5\rho U^2 D} \quad (2)$$

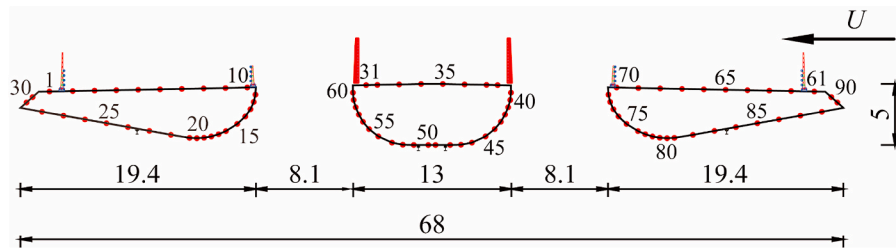
$$C_M(t) = \frac{\sum_{i=1}^n \{p_i(t) \Delta L_i [\cos(\alpha_i) x_i + \sin(\alpha_i) y_i]\}}{0.5\rho U^2 D^2} \quad (3)$$

Fig. 8 shows the VEF coefficient time-history at the extreme point and spectra of the four VIV stages. Spectra of pre-VIV and descent stages present “broadband” characteristics; there is no predominant frequency close to the natural frequency of the structure and the frequency is dispersed in the low-frequency region. The predominant frequencies of the ascent stage and extreme point are near to the structural natural frequency. The normalized amplitude of the extreme point is the largest. Upon entering the descent stage, the predominant frequency of VEF is 5.24 Hz with normalized amplitude of 0.065, which is clearly different from the natural frequency of the structure. At this time, the VIV amplitude drops sharply, indicating that the VIV response is closely related to the VEF amplitude and has clearly observable evolutionary characteristics. The spectrum of the extreme point has an obvious second harmonic of 6.75 Hz, which is twice the predominant frequency; thus, the VEF has obvious nonlinear characteristics. There is no second harmonic in the other three stages. The nonlinearity of VEF is also evolutionary.

During VIV, especially in the ascent stage and extreme point, vortex



a) Pressure measurement sections



b) The pressure tubes (unit: m)

Fig. 3. Pressure measurement sections and tubes.

shedding frequency is captured by structural vibration, so the VEF is dominated by structural vibration. The predominant frequencies of the displacement and VEF at the extreme point are almost equal, which differs significantly from cases without VIV. VEF shows obvious evolutionary characteristics.

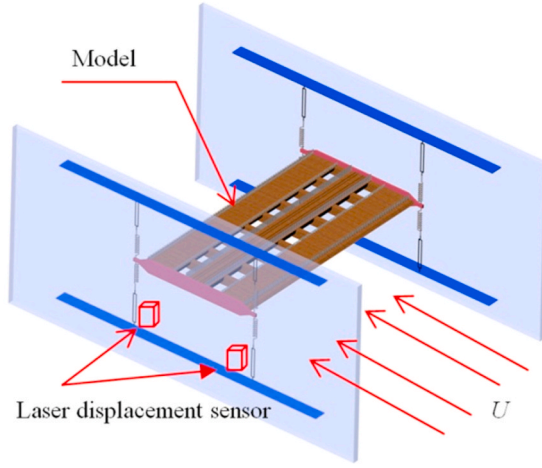
4.1. VEF components

The VEF components distribution of split box girders among boxes with and without VIV is worthy of attention. To determine the evolutionary characteristics of global VEF and its components during VIV, the pre-VIV and extreme point information were isolated as study objects; the mean and RMS values are shown in Fig. 9. In these two stages, the mean values of the railway and downstream boxes are negative while those of the upstream box are positive. The upstream and downstream boxes are the main components of the global VEF, and the upstream box is predominant between them. The VEF coefficient mean values of global, upstream, and railway boxes in the two stages are similar and those of the downstream box slightly increase at the extreme point. The upstream and downstream boxes are still the main components of the global VEF coefficient RMS values in both stages. Unlike the mean values, the downstream box accounts for more in terms of RMS value.

The global RMS values and those of the three boxes at the extreme point significantly increase, though to varying extent, compared with the pre-VIV stage – especially the global, upstream, and downstream boxes, which significantly differ from those without VIV.

4.2. VEF spanwise correlations

Turbulence intensity, angle of attack, and three-dimensional (3D) characteristics of the actual wind field, make the natural wind not completely synchronized along the bridge span. Wind loads occurring simultaneously at different locations along the bridge axis are not completely consistent. Previous researchers have mainly used conservative, two-dimensional (2D) VEF models (Sun et al., 2019). Usually, the correlation of wind speed is used to replace the correlation of VEF, however, correlation of the latter is significantly higher than that of the former (Nagao et al., 2003). The VEF spanwise correlation is the focus of the present study. According to the arrangement of the pressure measurement sections (Section 2), VEF correlations at six different spanwise distances were obtained by Eq. (4) and shown in Fig. 10a. The correlation coefficients of the four stages all decrease as the spanwise distance increases. The correlation coefficient in the pre-VIV stage is small, and gradually increases with VIV development until reaching its maximum



a) Test system



b) The DSM4000 Scanivalve system

Fig. 4. Test configuration diagram.

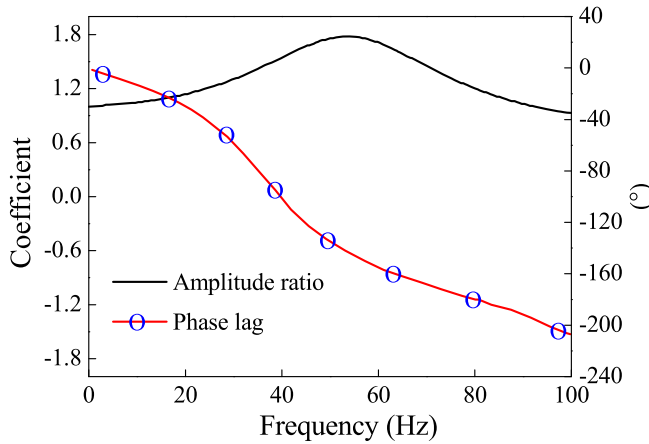


Fig. 5. Frequency response transfer function of tubing system.

at the extreme point. The correlation coefficient decreases sharply in the descent stage, to where it is smaller than in the pre-VIV stage. The correlation coefficients in pre-VIV, ascent, and descent stages are mainly dominated by spanwise distances, while they are determined by structural vibration at the extreme point, which almost breaks the limit of dimensionless spanwise distance.

$$\lambda_{ij} = \frac{E[F_{i-vortex} - \bar{F}_{i-vortex}]E[F_{j-vortex} - \bar{F}_{j-vortex}]}{D[F_{i-vortex}]D[F_{j-vortex}]} \quad (4)$$

Table 1
Test parameters.

Parameter	Description	Unit	Prototype	Model (1:30)
Geometric dimension	Length	m	/	3.90
	Width	m	68.0	2.27
	Height	m	5.0	0.17
Mass	Completion state	kg/m	74 937.0	83.26
	Completion state	kg.m ² /m	25 510	31.49
Mass moment of inertia	Completion state	m	093.0	
	Completion state	m	093.0	
Damping ratio	Vertical	/	3‰	1.92‰
	Torsional	/	3‰	2.54‰
Frequency	V-S-1	Hz	0.135 9	2.26
	T-S-1	Hz	0.253 1	3.39

*V: vertical; T: torsional; S: symmetrical; V-S-1 denotes first-order symmetrical vertical modal.

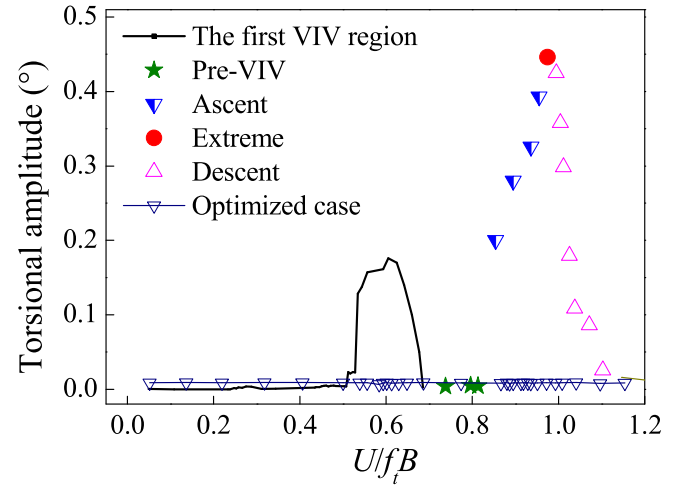


Fig. 6. VIV response and the stage division for analysis.

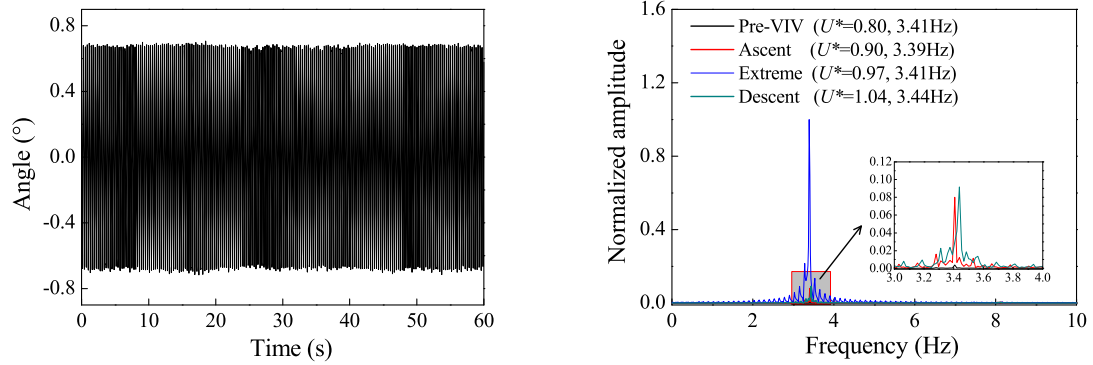
where λ_{ij} is the correlation coefficient between the i -th and j -th VEF. F_{vortex} is the global VEF calculated by Eq. (3). $E[\cdot]$ and $D[\cdot]$ denote mathematical expectation and variance, respectively.

In the 1960s, scholars investigated VEF spanwise correlations. It is believed that the VEF spanwise correlation decreases exponentially as spanwise distance increases:

$$R(x/D) = (1 - d_i) \exp[-Cx/D] + d_i \quad (5)$$

where R is the VEF spanwise correlation coefficient, C is the exponential decay coefficient, x/D is the dimensionless spanwise distance and d_i is the horizontal asymptote which represents the fully correlated part.

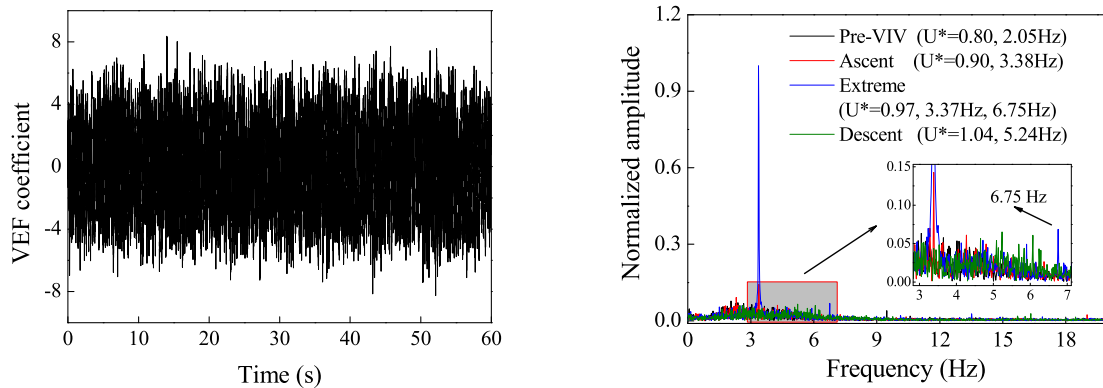
Fitting curves of VEF spanwise correlations in the four VIV stages are shown in Fig. 10b. The decay rate of the curve increases with decay coefficient C . As a more important parameter, d_i represents the lowest level of correlations. In pre-VIV and ascent stages, d_i is at a relatively low level and the values of C are also close. As VIV amplitude increases, d_i grows rapidly and reaches its maximum at the extreme point; C has also increased to 28.757, which indicates VEF spanwise correlations are at a relatively high level at this time. d_i decreases sharply in the descent stage or even smaller than that of the pre-VIV stage and C reaches its maximum, resulting in a sharp decrease in VIV amplitude. d_i and C , especially d_i , at the extreme point, are mainly controlled by structural vibration and are dominated by the spanwise distance in the other three stages.



a) Torsional angle time-history at the extreme point

b) Displacement spectra of four stages

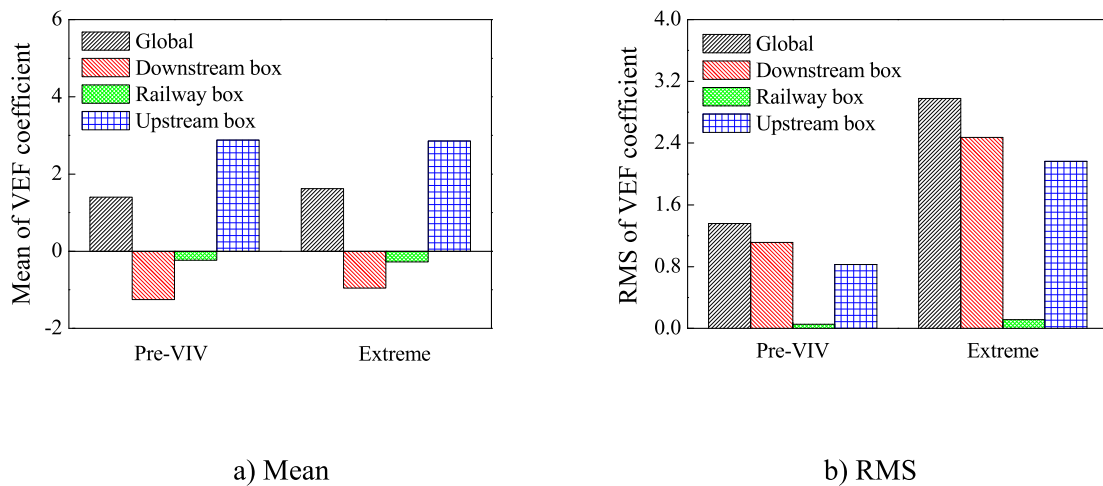
Fig. 7. Displacements and corresponding spectra.



a) VEF time-history at extreme point

b) VEF spectra of four stages

Fig. 8. VEF time-histories and corresponding spectra.



a) Mean

b) RMS

Fig. 9. Mean and RMS values of VEF and its components.

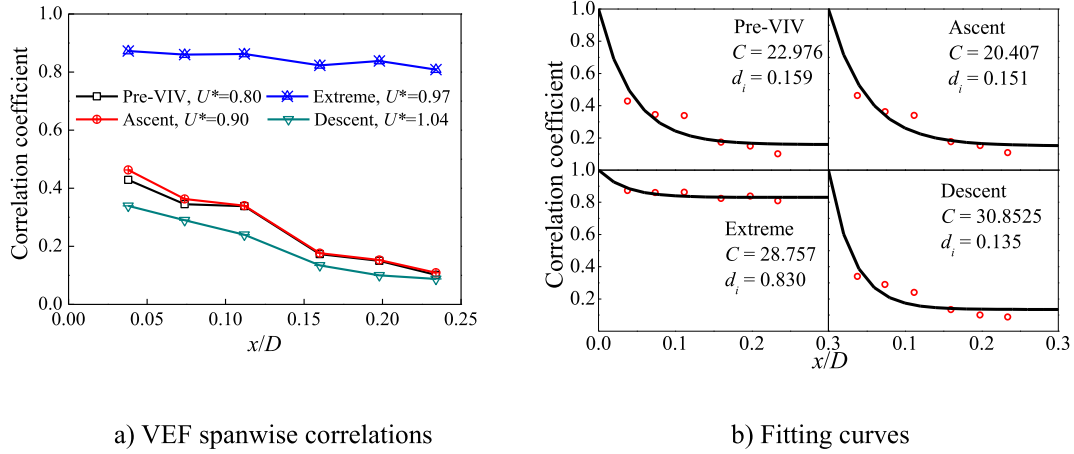


Fig. 10. VEF spanwise correlations and corresponding fitting results.

5. Wind pressure distribution

The spatial distribution of wind pressure coefficient represents the pressure distribution on the structure surfaces in a specific state and is unrelated to wind speed. The time-history of the wind pressure coefficient at the i -th tube can be defined as:

$$C_{p_i}(t) = \frac{p_i(t)}{1/2\rho U^2} \quad (6)$$

where $p_i(t)$ is the wind pressure time-history at the i -th tube; ρ is the air density (1.225 kg/m^3).

For convenience, the section surface is divided here into five typical areas numbered 1 to 5 in Fig. 11 and Table 2.

5.1. Mean and RMS values

Fig. 12a shows the spatial distribution of the mean values of wind pressure coefficient. The pressure coefficient at each tube is a straight line perpendicular to the surface through this point. The lines pointing to the inside and outside of the section indicate positive and negative values, namely, “impaction” and “suction”. The values are determined by the scale on the left. The mean distribution of the four stages is consistent and the value difference is very small. The windward of the upstream wind fairing shows positive pressures, while the other parts of the section show negative pressures. There are high and dense wind and safety barriers on the upper surface of the girder, the direct contact between airflow and section surface is prevented, so the entire upper surface is negative pressure and changes gently.

The airflow separates and sheds at the end of the ancillary facilities, moves backward, and merges into the wake. The mean distribution shows sudden changes on the lower surface, which are mainly located in areas 1 to 3. The wind fairing is a sharp bluff body under which separation bubbles are formed, causing a sudden change in pressure in area 1 (Schewe and Larsen, 1998). As the airflow passes through the maintenance tracks protruding from the lower surface of the girder, especially the upstream one, it forms vortex shedding behind it resulting in a sudden pressure change at the location of the maintenance tracks. The

Table 2

Description of divided surfaces.

Region	Description
Gap 1	Upstream and downstream curved surfaces in upstream gap
Gap 2	Upstream and downstream curved surfaces in downstream gap
Area 1	Lower sloping surface of upstream box
Area 2	Downstream curved surface in Gap 1
Area 3	Downstream curved surface in Gap 2
Area 4	Upper surface of downstream box
Area 5	The wake

webs of the boxes are transited by curved surfaces. A better streamlined curve is more sensitive to Re than a sharp corner and aggravates the viscous effect of the fluid (Larose and D’Auteuil, 2006). The gaps make the flow around the section more complicated, especially within the gaps, which is different from the integral box girder.

The mean distribution of the railway box shows that area 2 is markedly affected by the wake of the upstream box. The air flow through the upstream box forms an “impaction” effect on the downstream boxes. Upon reaching the extreme point, the mean values of areas 1 to 3 increase rapidly, indicating that the impaction effect intensifies as VIV amplitude increases. The spatial distribution of the mean values of wind pressure coefficient shows evolutionary characteristics in the VIV process.

The RMS values are all positive pointing toward the outside of the section and are also determined by the scale on the left. As shown in Fig. 12b, the RMS values on the lower surface of the section and the near wake changes significantly in different VIV stages. The RMS distribution is smooth and the value is the smallest in pre-VIV stage, at which point the model does not vibrate. When VIV occurs throughout ascent, extreme point, and descent stages, the RMS values in areas 1 to 5 increase rapidly with VIV amplitude then peak at the extreme point. The RMS in the descent stage decreases sharply with the decrease in VIV amplitude, showing obvious evolutionary characteristics.

The 4th, 44th, and 88th typical tubes in areas 4, 2, and 1 were selected to analyze the RMS evolutionary characteristics during VIV. The RMS values of the typical tubes and the VIV response are shown in Fig. 13.

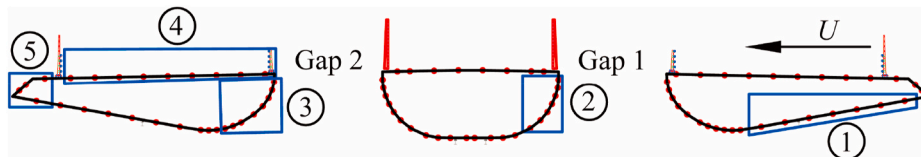


Fig. 11. Section surface division.

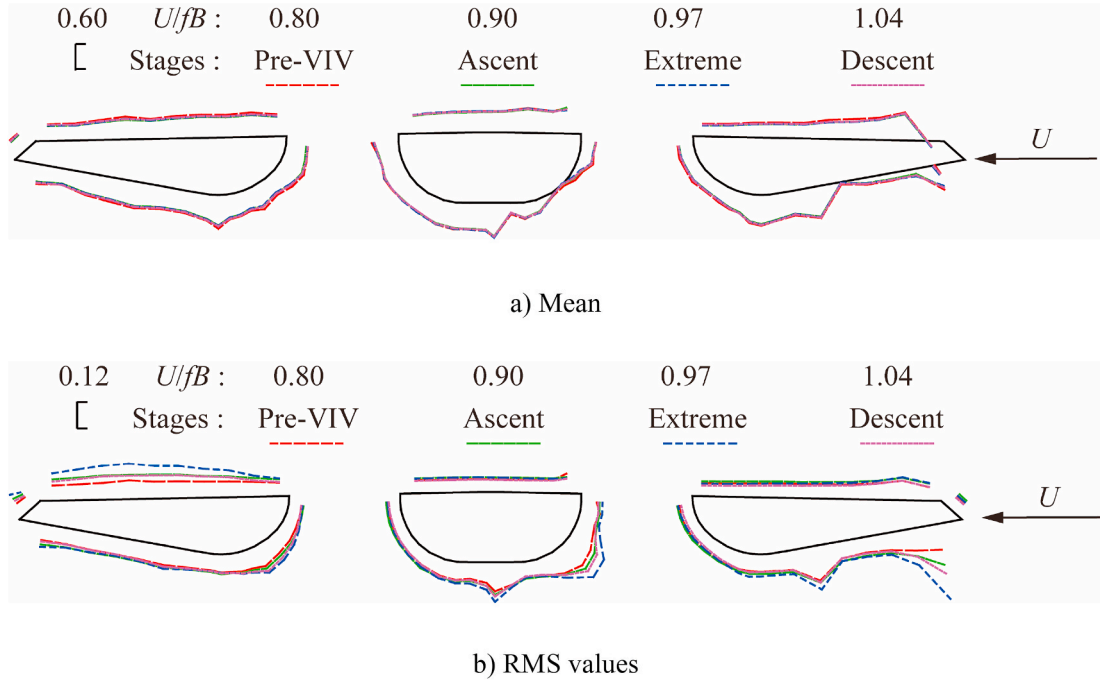


Fig. 12. Distribution of mean and RMS values of wind pressure coefficient.

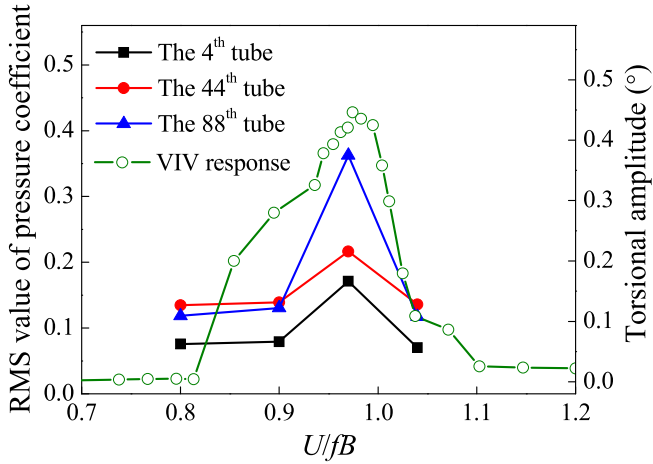


Fig. 13. RMS values of typical tubes and the VIV response.

The RMS values and the VIV response are in accordance – both increase alongside VIV development until peaking at the extreme point, and decrease sharply in the descent stage. Again, obvious evolutionary characteristics can be observed.

The evolutionary law of the mean and RMS values of the wind pressure coefficient in the VIV process indicate vortex formation and movement in areas 1 to 5, which cause changes in RMS values. The wake of the upstream box forms an impaction effect on the downstream boxes, which complicates the flow structure in the gaps and triggers periodic lift and moment, resulting in VIV. The separation bubbles below the upstream wind fairing and the violently fluctuating wake are also closely related to VIV.

5.2. Correlations between local aerodynamic forces and global VEF

The correlations between local aerodynamic forces and global VEF illustrate their relationship between different regions throughout VIV. The correlation coefficient γ_i between local aerodynamic force at the i -th

tube and the global VEF can be defined as:

$$\gamma_i = \frac{E \left[\left(F_{vortex} - \bar{F}_{vortex} \right) \left(p_i - \bar{p}_i \right) \right]}{D(F_{vortex})D(p_i)} \quad (7)$$

where F_{vortex} and $\bar{F}_{vortex}$ are the time-history and mean of the global VEF, respectively; p_i and $\bar{p}_i$ are the time-history and mean of aerodynamic force at the i -th tube, respectively. $E[\cdot]$ and $D[\cdot]$ denote mathematical expectation and variance, respectively. The correlation coefficient γ_i value lies in the interval $[-1, 1]$. $\gamma_i = -1$ indicates that the local aerodynamic force is completely negatively correlated with the global VEF, that is, it is completely reversed and has a complete suppression effect. $\gamma_i = 0$ means that there is no linear correlation, that is, neither promotion nor inhibition. $\gamma_i = 1$ indicates that the local aerodynamic force is completely positively correlated with the global VEF, that is, it is completely synchronized and there is complete promotion.

Fig. 14 shows the distribution of correlation coefficients between the local aerodynamic force at each tube and the global VEF in the VIV process. The lines on the inner and outer sides of the section represent negative and positive values, respectively. (The spatial distribution of contribution values is the same in all other representations.) In the pre-VIV stage, the correlation coefficients are relatively low. The correlation coefficients of the upper surface are positive for the upstream box, those of the lower surface are negative, and those at the wind fairing are positive. With the vertical center line of the railway box as the symmetry axis, the upstream and downstream correlation coefficients of the railway box have an asymmetrical distribution; the upstream ones are negative and the downstream ones are positive. For the downstream box, all are correlation coefficients positive except at the wind fairing.

The distribution of correlation coefficients in the ascent stage is consistent with that of the pre-VIV stage. The correlation coefficients at the wind fairing and lower surface of the upstream box, area 2, the upper surface of the railway box, and area 3 increased slightly. Upon reaching the extreme point, the spatial distribution of the correlation coefficients changes significantly – especially where the absolute values of the correlation coefficients at the upstream wind fairing, the upstream box surface, gap 1, and areas 1 to 5 increase sharply. Except for a small part of the lower surface of the upstream box and area 3, which show

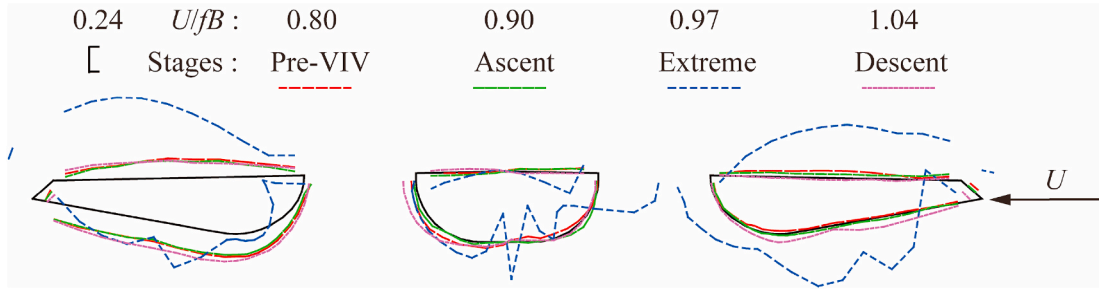


Fig. 14. Distribution of correlation coefficients between local aerodynamic forces and global VEF.

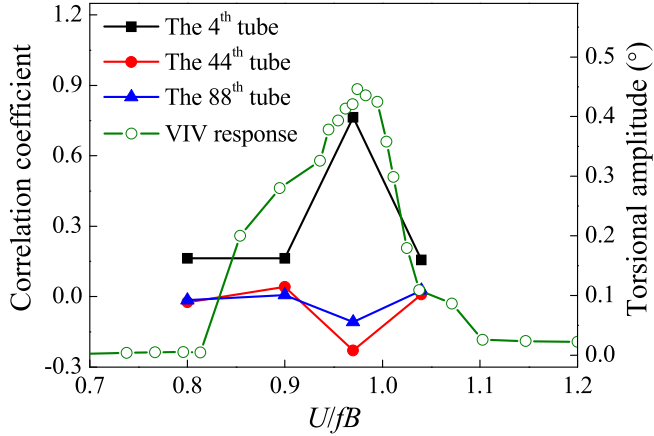


Fig. 15. Correlation coefficients of typical tubes and VIV response.

negative correlation coefficients, the others are all positive with large values. In the descent stage, the correlation coefficients are restored to the same distribution and values as that of the pre-VIV and ascent stages. The correlation coefficients between local aerodynamic forces and global VEF show obvious evolutionary characteristics during VIV.

To intuitively display the above-mentioned synchronous evolutionary characteristics, the 4th, 44th, and 88th tubes were isolated as shown in Fig. 15. The 4th tube is positively correlated with the global VEF while the 44th and 88th tubes show negative correlations. In the pre-VIV stage, the correlation coefficients of the three tubes are all at a low level. As VIV amplitude increases, the absolute values of correlation coefficients gradually increase from the ascent stage to the extreme point, where they peak. In the descent stage, the correlation coefficient distribution returns to a small level consistent with that of the pre-VIV stage.

The increase in correlation coefficients between local aerodynamic forces and global VEF on the upper and lower surfaces of the upstream box (except for partial lower surfaces of the wind fairing), gap 1, and areas 4 and 5 are likely the main reason for severe torsional VIV.

5.3. Contributions of local aerodynamic forces to global VEF

The values of local aerodynamic forces to global VEF can reveal the positive and negative contributions made by different areas of the section during VIV. The product of standard deviation in aerodynamic force at the i -th tube and the correlation coefficient between aerodynamic force at the i -th tube with global VEF is defined as the contribution value C_i made by the tube to the global VEF:

$$C_i = \gamma_i \sigma_i \quad (8)$$

where γ_i is the correlation coefficient between aerodynamic force at the i -th tube and global VEF; σ_i is the standard deviation of the aerodynamic force at the i -th tube.

The spatial distribution of contributions made by each tube to the global VEF is shown in Fig. 16. In the pre-VIV stage, the contribution of the entire section is very small since the model does not vibrate. The contributions of the upstream and railway boxes almost coincide with the section profile, though those of the downstream box are a little larger. In the ascent stage, the contribution of the upper and lower surfaces of the upstream box, especially the area near the wind fairing, tend to increase. The contributions of the railway and the downstream boxes do not change significantly at this point. Upon reaching the extreme point, the spatial distribution of the contribution values changes significantly. The contribution of the upstream and downstream boxes increase sharply, except for the negative contributions of the surface below the upstream wind fairing and area 3, which are all positive contributions with large values. In the descent stage, the distribution of contributions is restored but slightly exceeds that in pre-VIV and ascent stages. The contributions of local aerodynamic forces to global VEF show obvious evolutionary characteristics in the VIV process.

Fig. 17 shows the contribution values of the 4th, 44th, and 88th typical tubes to global VEF with VIV development. Similar to the correlation coefficient between local aerodynamic forces and the global VEF, the contribution value of the 4th tube is positive while those of the 44th and 88th tubes are negative. In pre-VIV and ascent stages, with a small VIV amplitude, the contributions are very small. As the VIV amplitude increases to the extreme point, both positive and negative contribution

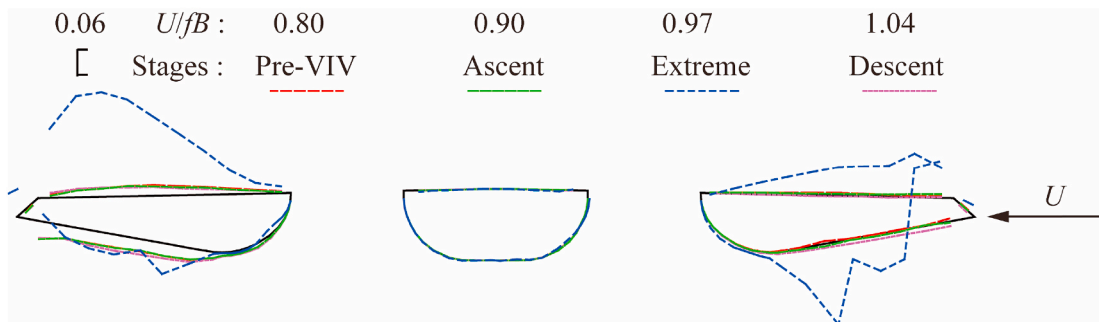


Fig. 16. Distribution of contributions made by local aerodynamic forces to global VEF.

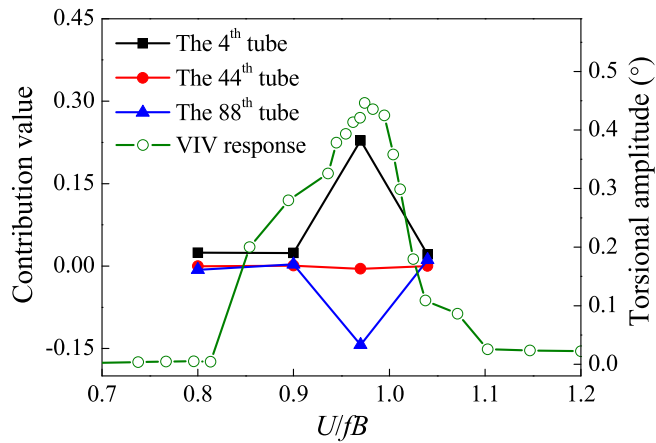


Fig. 17. Contributions of typical tubes and VIV response.

values reach their maxima. The values decreased to a lower level in the descent stage, showing evolutionary characteristics.

The aerodynamic forces on the upstream box surface (except partial lower surfaces of the wind fairing), area 4, and area 5 make the main contribution to the global VEF during VIV, which is the main cause of torsional VIV.

5.4. Phase lags between local aerodynamic forces and global VEF

Phase lags between two time-series at a certain frequency reflect the spatial relationship between them over a range from -180° to 180° . Phase lags indicate relative locations between aerodynamic forces at each tube and the global VEF during VIV, which further indicate VIV inducements. The phase lag is 0° when the correlation coefficient is 1, where the phase is the same. The phase lag is 90° when the correlation coefficient is 0, which indicates orthogonality. The phase lag is 180°

when the correlation coefficient is -1 , indicating inversion. Phase lags between the aerodynamic force of each tube and global VEF at the predominant frequency were corrected based on Fig. 5 as shown in Fig. 18.

In the pre-VIV stage, the phase lag curve is scattered with many inflection points and shows obvious local features. At this point, the relationship between aerodynamic forces of adjacent tubes is weak and the global VEF has a small amplitude. In the ascent stage, the phase lag curve becomes smooth and takes a shape almost opposite to that of the pre-VIV stage, especially in upstream and railway boxes. The connection between adjacent tubes becomes tighter and most phase lags fall between 0° and 90° .

At the extreme point, the phase lag curve is still smooth and tends to decrease on the whole with values between -90° and 90° . Phase lags of the upper and lower surfaces of the upstream box (except partial lower surfaces of the wind fairing) and areas 4 to 5 range between 0° and 90° , which promotes the global VEF. In the descent stage, similar to the pre-VIV stage, the phase lags are further reduced with values within 0° to 90° and the local features are restored. The interconnection between the tubes rapidly weakens and the VEF and VIV amplitude both rapidly decrease. From the perspective of phase lag, aerodynamic forces on the upper and lower surfaces of the upstream box (except for partial lower surfaces of the wind fairing) and areas 4 to 5 play a major role in promoting the global VEF, which is closely related to VIV. Phase lags between local aerodynamic forces and the global VEF also show obvious evolutionary characteristics in the VIV process.

6. Conclusions

A large-scale section model displacement and pressure measurement wind tunnel test was carried out on a split three-box girder bridge in this study by free vibration method. Synchronous measurements of displacement and pressure were used to obtain the time-histories of pressure at each tube and displacements in the VIV process, which was divided into four stages: pre-VIV, ascent, extreme point, and descent.

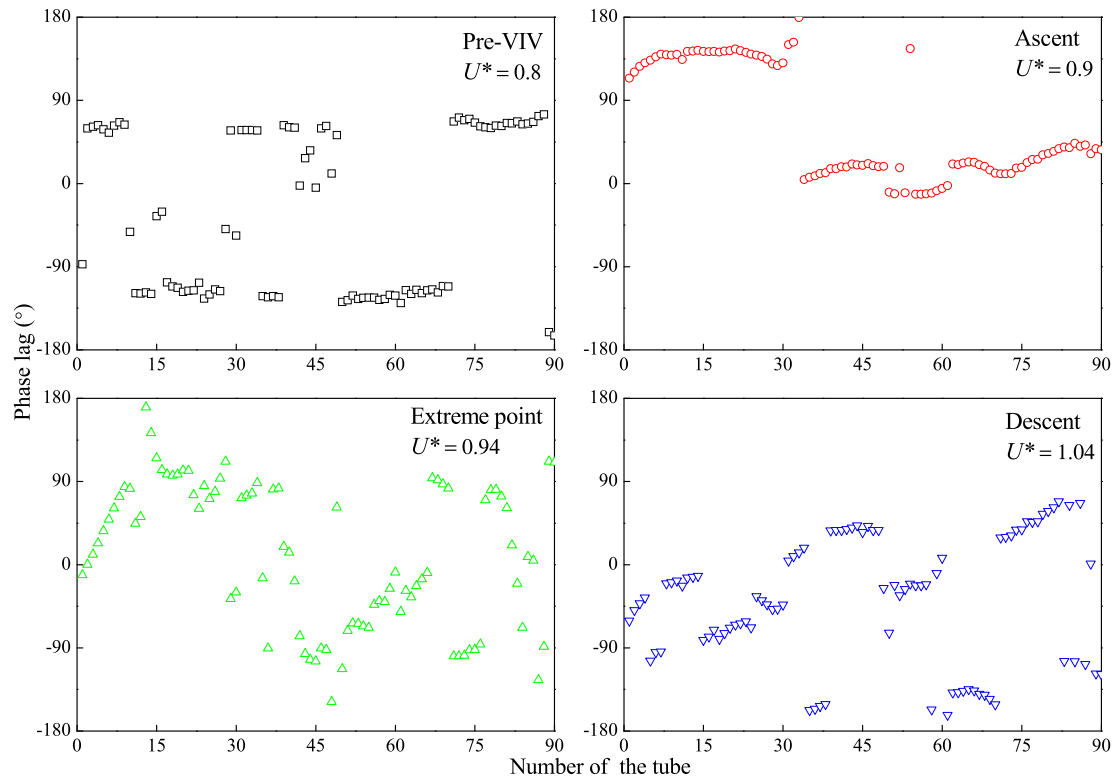


Fig. 18. Phase lags between local aerodynamic forces and global VEF.

The global VEF was determined by integrating the tube pressures. Based on the data above, the evolutionary characteristics of typical VEF indexes throughout the VIV process were analyzed. The inducement mechanism of torsional VIV in the split three-box girder was determined as well. This work may illuminate the aerodynamic behavior and VIV control measures of novel split three-box girder bridges. Some conclusions can be summarized as follows.

- 1) Spectrum analysis shows that both displacement and VEF have obvious evolutionary characteristics in the VIV process. High-order harmonics can be observed in the VEF spectrum at the extreme point, which indicates nonlinear characteristics of VIV.
- 2) VEF components between different boxes and the spatial distribution of mean and RMS wind pressure coefficient values indicate that vortex formation and movement in areas 1 to 5. An increase in the RMS values in these areas is the main cause of severe torsional VIV.
- 3) The spanwise correlation shows evolutionary characteristics during VIV. It is mainly dominated by spanwise distance in pre-VIV, ascent, and descent stages but is controlled by structural vibration at the extreme point.
- 4) The correlation between local aerodynamic forces and the global VEF, and the contributions of local aerodynamic forces to the global VEF, show evolutionary characteristics in the VIV process. The contributions of area 3 and partial surfaces of area 1 are negative, but the positive correlation coefficients and contributions of larger values on the upstream box surface (except for partial lower surfaces of the wind fairing), gap 1, area 4, and area 5 far exceed the negative ones, resulting in torsional VIV with large amplitude.
- 5) The spatial distribution of phase lags between local aerodynamic forces and the global VEF in the ascent stage and extreme point differ markedly from that in pre-VIV and descent stages. In pre-VIV and descent stages, the phase lag curve is scattered with many inflection points and shows obvious local features, while the curve becomes smooth at the ascent stage and extreme point. The connection between adjacent tubes tightens, resulting in large-amplitude VEF which leads to VIV.
- 6) In general, the evolution of the typical VEF indexes on the upstream box surface (except for partial lower surfaces of the wind fairing), gap 1, area 4, and area 5 is primarily responsible for the torsional VIV of split three-box girder bridges.

CRedit authorship contribution statement

Fengfan Yang: Conceptualization, Methodology, Software, Writing – original draft, Supervision. **Shixiong Zheng:** Resources, Writing – review & editing, Funding acquisition. **Qiang Zhou:** Validation, Investigation. **Zhengxi Yan:** Visualization, Investigation. **Zihao Ding:** Investigation. **Xueyang Tai:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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